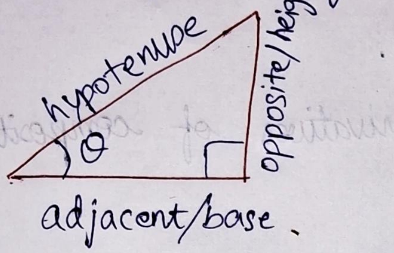
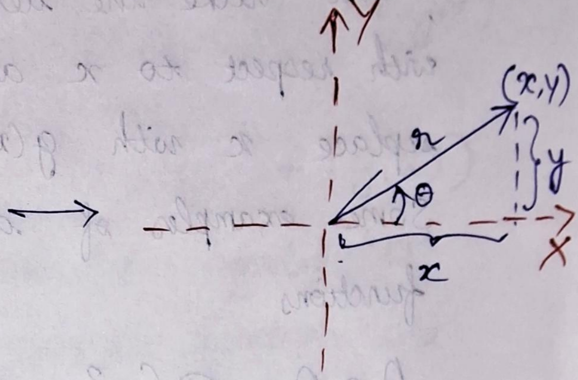
03rd April, 2026 (Fri)

Homework on pre-requisites for Quantum Structures

Trigonometric





$$\sin\theta := \frac{\text{height}}{\text{hypotenuse}} \Rightarrow \text{height} = \text{hypotenuse}\cdot\sin\theta$$

$$\Rightarrow y = r\sin\theta$$

$$\cos\theta := \frac{\text{base}}{\text{hypotenuse}} \Rightarrow \text{base} = \text{hypotenuse}\cdot\cos\theta$$

$$\Rightarrow x = r\cos\theta$$

$$\tan\theta := \frac{\sin\theta}{\cos\theta}$$

Pythagoras theorem:

$$\text{hypotenuse}^2 = \text{base}^2 + \text{height}^2$$

$$\Rightarrow r^2 = r^2(\cos^2\theta + \sin^2\theta)$$

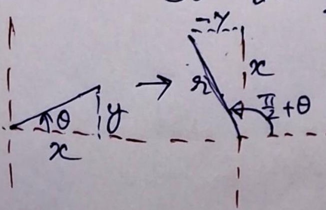
$$\Rightarrow \cos^2\theta + \sin^2\theta = 1$$

Decomposition of angles

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

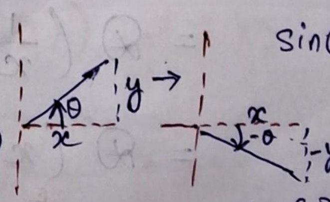
$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

so $\theta \to \frac{\pi}{2}+\theta$ $\qquad\qquad$ $\theta \to -\theta$



$$\sin\left(\frac{\pi}{2}+\theta\right) = \frac{x}{r} = \cos\theta$$

$$\cos\left(\frac{\pi}{2}+\theta\right) = \frac{-y}{r} = -\sin\theta$$



$$\sin(-\theta) = \frac{-y}{r} = -\sin\theta$$

$$\cos(-\theta) = \frac{x}{r} = \cos\theta$$

Question

Show that $\cos^2\theta = \dfrac{1+\cos 2\theta}{2}$

$\sin^2\theta = \dfrac{1-\cos 2\theta}{2}$

Differentiation - Integration duality

Given two functions $f(x)$ and $g(x)$, and differentiation operation $\mathcal{D}$ is defined as

$$\mathcal{D} := \frac{d}{dx}, \Rightarrow \mathcal{D}(f) = \frac{df}{dx}, \text{ and}$$

integration operation $\mathcal{I}$ is defined as

$$\mathcal{I} := \int dx \Rightarrow \mathcal{I}(g) = \int g(x)\,dx.$$

~~then~~ If differentiation of $f(x)$ gives $g(x)$, then

integration of $g(x)$ gives $f(x)$!

$$\mathcal{D}(f) = g \Rightarrow \mathcal{I}(g) = f$$

i.e. $\dfrac{d}{dx} f(x) = g(x) \Rightarrow \int g(x)\,dx = f(x) + C$

Differentiation of IL$\boxed{A}\boxed{T}\boxed{E}$[2] functions

A : $\mathcal{D}(x^n) = \dfrac{d}{dx} x^n = n x^{n-1}$

T : $\mathcal{D}(\sin x) = \dfrac{d}{dx} \sin x = \cos x$

$\mathcal{D}(\cos x) = \dfrac{d}{dx} \cos x = -\sin x$

E : $\mathcal{D}(e^x) = \dfrac{d}{dx} e^x = e^x$

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1 upto an integration constant c.

2 I (Inverse Trigonometric) L Logarithm A (Arithmetic) T (Trigo) E (Exponential)

Chain rule of differentiation of composite functions

Given two functions $f(x)$ and $g(x)$, a composite function of $f(x)$ and $g(x)$ is

$$h(x) := f\circ g\,(x) := f(g(x))$$

The composite operation has symbol '$\circ$'.

Some examples of composite functions are

$A \circ A$: $\quad x^2 \circ x^2 := x^2(x^2) = (x^2)^2 = x^4$

$A \circ T$: $\quad x^3 \circ \sin x := x^3(\sin x) = \sin^3 x$

~~A ∘ E :~~

$E \circ A$: $\quad e^{-x} \circ x^2 := e^{-x}(x^2) = e^{-x^2}$

Now,

differentiation of a composite function is given by the chain rule

$$\mathcal{D}(f\circ g) := \frac{d}{dx}(f\circ g)(x)$$
$$= \frac{d}{dx} f(g(x))$$
$$= \frac{d}{dg(x)} f(g(x)) \cdot \frac{d}{dx} g(x)$$

$$\boxed{\mathcal{D}(f\circ g) = \left.\frac{d}{dx} f(x)\right|_{g(x)} \left.\frac{d}{dx} g(x)\right|_{x}}$$

Here, the symbol

$$\frac{d}{dx} f(x)\Big|_{g(x)}$$

means take the derivative of $f(x)$ with respect to x and later replace x with $g(x)$.

Some examples of derivatives of composite functions

$A \circ A$: $\mathcal{D}(x^2 \circ x^2) = \frac{d}{dx} x^2\Big|_{x^2} \cdot \frac{d}{dx} x^2\Big|_{x}$

$= 2x^{2-1}\Big|_{x^2} \cdot 2x^{2-1}\Big|_{x}$

$= 2x\Big|_{x^2} \cdot 2x\Big|_{x}$

$= 2x^2 \cdot 2x$

$= 4x^3$

$\downarrow$

$\mathcal{D}(x^4)$

$= \frac{d}{dx} x^4$

$= 4x^{4-1}$

$= 4x^3$

$A \circ T$: $\mathcal{D}(x^3 \circ \sin x) = \frac{d}{dx} x^3\Big|_{\sin x} \cdot \frac{d}{dx} \sin x\Big|_{x}$

$= 3x^2\Big|_{\sin x} \cdot \cos x\Big|_{x}$

$= 3\sin^2 x \cdot \cos x$

$E \circ T$: $\mathcal{D}(e^{-x} \circ x^2)$

$= \mathcal{D}\left(\frac{1}{e^x} \circ x^2\right)$

$= \mathcal{D}\left(\frac{1}{x} \circ e^x \circ x^2\right)$

$= \mathcal{D}(x^{-1} \circ e^x \circ x^2)$

$\quad f \quad\quad g \quad\quad h$

$$= \cancel{\frac{d}{dx} x^{-1} \Big|} = \mathcal{D}(\underbrace{x^{-1}}_{f} \circ \underbrace{e^{x^2}}_{g})$$

$$= \frac{d}{dx} x^{-1} \Big|_{e^{x^2}} \cdot \frac{d}{dx} e^{x^2} \Big|_{x}$$

$$= -1 \cdot x^{-2} \Big|_{e^{x^2}} \cdot \mathcal{D}(e^{x^2})$$

$$= -(e^{x^2})^{-2} \cdot \mathcal{D}(e^{x} \circ x^2)$$

$$= -e^{-2x^2} \left[\frac{d}{dx} e^{x} \Big|_{x^2} \cdot \frac{d}{dx} x^2 \Big|_{x} \right]$$

$$= -e^{-2x^2} \cdot e^{x} \Big|_{x^2} \cdot 2x \Big|_{x}$$

$$= -e^{-2x^2} \cdot e^{x^2} \cdot 2x$$

$$= -2x e^{-x^2}$$

Integration of IL$\boxed{A}\boxed{T}\boxed{E}$ functions

A : $\mathcal{I}(x^n) = \int x^n \, dx = \frac{x^{n+1}}{n+1} + c$

T : $\mathcal{I}(\sin x) = \int \sin x \, dx = -\cos x + c$

$\mathcal{I}(\cos x) = \int \cos x \, dx = \sin x + c$

E : $\mathcal{I}(e^x) = \int e^x \, dx = e^x + c$

Integration of composite functions

NOTE: Integration of composite functions is an

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Questions

Find $\int x^4 \, dx$, $\int \sin^3 x \, dx$, $\int e^{-x^2} dx$